

Amivantamab (Rybrevant)

A triple-action bispecific antibody targeting EGFR and MET — and the trial results that made headlines today

Drug Amivantamab-vmjw **Brand** Rybrevant / Rybrevant Faspro **Developer** Johnson & Johnson (Janssen)

FDA Approval May 21, 2021 **Prepared** May 30, 2026

Sources The Guardian · J&J Press Releases · PubMed · ClinicalTrials.gov · ASCO 2026

TODAY'S RESULTS	MECHANISM	REGULATORY	TRIALS	NEXT STEPS
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Today at ASCO 2026, doctors reported that amivantamab **eradicated tumours entirely** in 15 patients whose cancer had spread and failed prior treatments. An international trial spanning 11 countries showed tumour shrinkage in more than a third of patients, with responses appearing **within weeks**. The results were called "unprecedented" by the investigators.

TODAY'S TRIAL RESULTS — ASCO 2026

The results reported today are from the **OrigAMI-4 study** — a Phase 1b/2 trial evaluating amivantamab in head and neck squamous cell carcinoma (HNSCC), a cancer type with very limited treatment options once standard therapies fail. These patients had already received checkpoint inhibitor immunotherapy and platinum-based chemotherapy without lasting benefit.

COMPLETE RESPONSES 15 Tumours eradicated entirely	OVERALL RESPONSE RATE >33% Tumour shrinkage in 1 in 3	MEDIAN OVERALL SURVIVAL 12.5 mo Despite poor-prognosis population	COUNTRIES 11 International trial
RESPONSE SPEED Weeks Median time to first response ~6 weeks	CLINICAL BENEFIT RATE 76% Response or durable stable disease		

In oncology, a **complete response** means no detectable tumour remaining on imaging. In patients with metastatic disease that has failed multiple prior lines of therapy, this outcome is rare enough to be genuinely unusual. Most drugs in this setting aim to slow progression — not eliminate disease. Getting 15 complete responses in a heavily pre-treated population is the number that justifies the word "unprecedented."

"This is a group of patients for whom treatment options are extremely limited, so seeing this level of benefit is very striking."

— Prof. Kevin Harrington, Consultant Oncologist, Royal Marsden NHS Foundation Trust

① Source: The Guardian, Andrew Gregory, May 30, 2026; Irish Times (same story); J&J OrigAMI-4 press release; ScienceDirect published results (NCT06385080)

WHAT AMIVANTAMAB IS

Amivantamab is a **fully human IgG1-based bispecific antibody** — an engineered protein that simultaneously targets two receptors on the surface of cancer cells: EGFR (epidermal growth factor receptor) and MET. Unlike conventional cancer drugs that hit one target, amivantamab attacks through three distinct mechanisms at once.

01

RECEPTOR BLOCKADE

Prevents growth signals from binding to EGFR and MET on the tumour cell surface, cutting off the fuel that drives tumour growth and division.

02

RECEPTOR DEGRADATION

Forces EGFR and MET to be internalised and destroyed by the cancer cell itself, reducing the number of active receptor targets available.

03

IMMUNE ACTIVATION

Recruits natural killer cells and macrophages to attack the tumour directly — antibody-dependent cellular cytotoxicity (ADCC) and trogocytosis.

The dual-target design matters because cancer cells frequently develop resistance by switching from one pathway to the other. When EGFR-targeted drugs lose effectiveness, tumours often activate MET to escape. Amivantamab blocks both simultaneously, making this escape route harder to use.

WHY EGFR AND MET TOGETHER

EGFR overexpression or mutation drives growth in a wide range of cancers — lung, colorectal, head and neck, and others. MET is the primary escape mechanism cancer cells use when EGFR-directed therapy stops working. Blocking both removes the most common resistance pathway at the same time as the primary target, which is why the drug works in patients who have already progressed on other EGFR-directed therapies.

① Source: ClinicalTrials.gov NCT05653427 Investigator's Brochure; Pharmacy Times clinical review, 2026; PubMed PMID 41807198

REGULATORY HISTORY & FDA APPROVALS

- May 21, 2021
FDA Accelerated Approval — NSCLC with EGFR Exon 20 insertion mutations, after progression on platinum-based chemotherapy. Based on CHRYSALIS Phase 1 trial data.
- 2023
MARIPOSA-2 Phase III results — amivantamab + chemotherapy ± lazertinib improved progression-free survival vs chemotherapy alone after osimertinib failure. First study to demonstrate PFS benefit in this setting.
- January 2026
Colorectal cancer data — OrigAMI-1 Phase 1b/2 showed durable responses in RAS/BRAF wild-type metastatic colorectal cancer in combination with FOLFOX/FOLFIRI chemotherapy. Phase 3 ongoing.
- February 18, 2026
FDA Breakthrough Therapy Designation — subcutaneous formulation (Rybrevant Faspro) for HPV-unrelated advanced head and neck squamous cell carcinoma after progression on platinum and checkpoint inhibitor.
- May 30, 2026
ASCO 2026 — OrigAMI-4 results — 15 complete tumour eradications, >33% response rate in heavily pre-treated HNSCC. Results presented in Chicago. Reported today in The Guardian.

① Source: FDA approval records; J&J press releases Jan 2026, Feb 2026; Annals of Oncology MARIPOSA-2; CancerNetwork ASCO 2026 coverage

ACTIVE CLINICAL PROGRAMME

Amivantamab is currently being evaluated in approximately **60 clinical trials** spanning multiple cancer types. The programme has expanded significantly from its original NSCLC focus.

STUDY	CANCER TYPE	SETTING	KEY RESULT	STATUS
CHRYSLIS-2 NCT04077463	NSCLC (atypical EGFR mutations)	First-line + lazertinib	Median OS 41.0 months; 55% alive at 36 months	Phase 1/1b — updated results ASCO 2026
MARIPOSA-2 NCT04988295	NSCLC (EGFR-mutated)	After osimertinib failure	First study to show improved PFS vs chemo in this setting	Phase III — published
OrigAMI-1 NCT05379595	Metastatic colorectal cancer	RAS/BRAF wild-type + chemo	Durable responses including liver metastases	Phase 1b/2 — Phase 3 ongoing
OrigAMI-4 NCT06385080	Head & neck (HNSCC, HPV-unrelated)	After checkpoint + platinum failure	45% ORR; 15 complete responses; 76% clinical benefit rate	Phase 1b/2 — results at ASCO 2026 today
OrigAMI-5 NCT07276399	Head & neck (HNSCC, HPV-unrelated)	First-line + pembrolizumab + carboplatin	Ongoing	Phase 3 — enrolling

① Source: ClinicalTrials.gov; CancerNetwork ASCO 2026 coverage; J&J press releases; OncLive

THE PATIENT STORY

One of the first patients to benefit from the OrigAMI-4 trial was **Carl Walsh, 56**, who was diagnosed with tongue cancer in May 2024 and joined the trial at the Royal Marsden in July 2025. His case is representative of the population the trial targets — a patient who had exhausted other options and for whom the available standard of care offered limited benefit.

CLINICAL CONTEXT

Head and neck squamous cell carcinoma is the cancer type in today's trial results. In the HPV-unrelated subtype specifically — the population studied in OrigAMI-4 — **EGFR and MET are overexpressed in 80–90% of tumours**, making amivantamab's dual-target mechanism directly relevant to the biology of the disease. Prior to amivantamab, single-agent chemotherapy in patients who had progressed on checkpoint inhibitors and platinum offered response rates in the single digits.

① Source: Irish Times/Guardian, May 30, 2026; Annals of Oncology OrigAMI-4 (NCT06385080); Pharmacy Times clinical review

Amivantamab was originally developed as an **intravenous infusion**, administered over several hours. The initial doses were split across two days due to the risk of infusion-related reactions. Premedication with corticosteroids, antihistamines, and antipyretics is standard.

A **subcutaneous formulation** (Rybrevant Faspro, with hyaluronidase) has been developed to improve patient convenience and reduce infusion-related reactions. The subcutaneous version was used in the OrigAMI-4 trial. In the trial data, administration-related reactions were reported in only 7% of participants — a substantial improvement over the intravenous formulation's infusion reaction profile.

① Source: PubMed PMID 41807198 (PALOMA study); Pharmacy Times clinical review; ScienceDirect OrigAMI-4 published results

WHAT HAPPENS NEXT

Today's results at ASCO 2026 are Phase 1b/2 data — meaningful signal, but not the definitive Phase 3 evidence that drives regulatory approval. The path forward for HNSCC specifically:

REGULATORY PATHWAY — HNSCC

The **FDA Breakthrough Therapy Designation** granted in February 2026 accelerates the regulatory dialogue with J&J and signals the agency views the Phase 1b/2 data as preliminary evidence of substantial improvement over existing therapies. The **OrigAMI-5 Phase 3 trial** (NCT07276399), combining subcutaneous amivantamab with pembrolizumab and carboplatin in first-line HNSCC, is currently enrolling. Phase 3 results will determine whether full FDA approval follows. The durability of the complete responses reported today — do those 15 patients remain tumour-free at 12 and 24 months — is the critical data point to watch.

Separately, amivantamab's colorectal cancer programme (OrigAMI-1) is advancing into Phase 3 first- and second-line trials. The lung cancer programme has already produced Phase 3 data. The breadth of the programme — 60 active trials — reflects J&J's view that the EGFR/MET dual target is applicable across a wide range of solid tumours where these receptors are overexpressed or mutated.

① Source: J&J Feb 18 2026 press release; CancerNetwork Breakthrough Designation coverage; ClinicalTrials.gov NCT07276399

SOURCES

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Chen MF et al., J Thorac Oncol 2026;21(3):103505 – Amivantamab + lazertinib, brain/leptomeningeal mets
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MARIPOSA-2, Annals of Oncology, Oct 2023 – Phase III post-osimertinib NSCLC
FDA Approval Record, May 21, 2021 – Accelerated approval, NSCLC Exon 20 insertion
ClinicalTrials.gov NCT04077463, NCT04988295, NCT05379595, NCT06385080, NCT07276399

Amivantamab Drug Intelligence
Brief · May 30, 2026

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Sources: SEC EDGAR, PubMed,
ClinicalTrials.gov, J&J

AMIVANTAMAB DRUG INTELLIGENCE BRIEF

Data: The Guardian · J&J Press
Releases
PubMed · ClinicalTrials.gov · ASCO
2026

Not medical advice
Deterministic · No AI summaries

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Sources: Andrew Gregory, The Guardian, May 30 2026 · J&J Press Releases Jan-Feb 2026 · NCT06385080, NCT04077463, NCT04988295,
NCT05379595, NCT07276399 · ASCO 2026 Annual Meeting presentations